

Unit 4 :: Special Theory of Relativity

☞ **PHY0400104: Classical Mechanics** ☞

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Syllabus

Unit 4: Special theory of relativity

Inadequacy of Galilean transformation; postulates of special relativity; Lorentz transformation; simultaneity and order of events; length contraction and time dilation; relativistic addition of velocities; variation of mass with velocity and mass-energy equivalence. Lorentz transformation as a rotation in spacetime; relation between proper time and coordinate time; relativistic kinematics: energy-momentum relation.

1 Equilibrium and potential energy

2 Theory of small oscillations

3 Physical problems and applications